

KICS Website

Complete Documentation

Al-Khwarizmi Institute of Computer Science
University of Engineering & Technology, Lahore

Version 2.0.0 | May 2026

KICS Website Documentation

Website: Al-Khwarizmi Institute of Computer Science (KICS)
Institution: University of Engineering & Technology, Lahore
URL: <https://kics-frontend-modern.vercel.app>
Version: 2.0.0
Last Updated: May 2026

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Technology Stack

Frontend Framework

Build Tools

SEO & Meta

- Structured data with JSON-LD

Deployment

- CDN: Global edge network

Color System & Design Theme

Primary Color Palette (Blue Tech Theme)

Color Name	Hex Code	Usage
----- ----- -----		
Primary 50	`#eff6ff`	Very light blue mist backgrounds
Primary 100	`#dbeafe`	Light blue elements
Primary 200	`#bfdbfe`	Soft blue tints
Primary 300	`#93c5fd`	Light tech blue accents
Primary 400	`#60a5fa`	Medium blue highlights
Primary 500	`#3b82f6`	Main interactive blue
Primary 600	`#2563eb`	Deep blue (primary CTA)
Primary 700	`#1d4ed8`	Rich blue hover states
Primary 800	`#1e40af`	Dark tech blue
Primary 900	`#1e3a8a`	Deep navy blue backgrounds
Primary 950	`#0b2447`	Deepest tech navy (dark sections)

Accent Colors (Cyan/Teal)

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Color Name	Hex Code	Usage
----- ----- -----		
Cyan 400	`#22d3ee`	Neon cyan highlights
Cyan 500	`#06b6d4`	Main cyan accent
Cyan 600	`#0891b2`	Deep teal

Neutral Colors (Slate)

Color Name	Hex Code	Usage
----- ----- -----		
Slate 50	`#f8f9fc`	Light backgrounds
Slate 100	`#f1f5f9`	Card backgrounds
Slate 500	`#64748b`	Secondary text
Slate 700	`#334155`	Body text
Slate 900	`#0f172a`	Headings, dark text

Background Gradients

Hero Gradient: `linear-gradient(135deg, rgba(11,36,71,0.9), rgba(30,64,175,0.85))`

Shadow System

Card: `0 4px 24px rgba(37,99,235,0.08)`

Page Structure & Layout

Overall Layout Pattern

Design Strategy

All Sections (15 Total)

1. Hero Section

- Purpose: First impression, main value proposition
- Layout: Full-screen image slider with text overlay
- Background: Gradient overlay on university photos
- Content:
- 4 rotating slides (auto-play every 5 seconds)
- Features:
- Glassmorphism text container

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Colors:

- Text: White

2. Announcements Ticker

Purpose: Display latest news and updates

Layout: Horizontal scrolling marquee

Background: Dark navy (#1e3a8a)

Animation: Infinite scroll (40s duration), pauses on hover

Content:

- 8 latest announcements (duplicated for seamless loop)

Features:

- Auto-scrolling left-to-right

Colors:

- Background: `primary-900` (#1e3a8a)

3. Video Section

Purpose: Showcase KICS overview video

Layout: Centered video player with heading

Background: White with grid pattern overlay

Content:

- Section heading: "Innovation in Action"

Features:

- YouTube/Vimeo embed support

Colors:

- Background: White

4. Stats Bar (KICS at a Glance)

Purpose: Showcase institutional achievements

Background: Dark navy with dot pattern

Content (6 Stats):

- 22+ Years of Excellence

Features:

- Animated counter (counts up on scroll into view)

Colors:

- Background: `primary-900` (#1e3a8a)

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Bottom Banner:

- IEEE Technical Collaborator

5. Achievement Badges

Purpose: Display awards and recognition

Background: Dark gradient with wave divider top

Content (4 Achievements):

- IEEE Gold Award (Best Research Paper 2024)

Features:

- Emoji icons in gradient boxes

Colors:

- Cards: White 10% opacity with backdrop blur

6. About Strip

Purpose: Introduce KICS mission and history

Layout: 2-column split (text left, feature cards right)

Content:

- Left: Al-Khwarizmi namesake history, KICS founding (2002)

Features:

- Scroll reveal animations (left/right)

Colors:

- Background: White/Slate 50

7. Research Section

Purpose: Showcase research areas and labs

Layout: 3-column grid with image cards

Content (6 Research Areas):

- AI & Machine Learning

Features:

- Image cards with hover zoom

Colors:

- Background: Gradient blue/cyan 50 shades

8. Featured Projects

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Purpose: Highlight student/startup projects

Layout: 3-column grid with image overlay cards

Background: Dark gradient with glowing orbs

Content (3 Projects):

- AquaTech (NVC 2026 Winner, \$20K)

Features:

- Image with award badge

Colors:

- Background: Dark navy gradient

9. Testimonials Section

Purpose: Alumni success stories

Layout: 3-column grid with quote cards

Background: Light gray with giant quotation marks

Content (3 Testimonials):

- Dr. Ali Hassan (MIT Media Lab)

Features:

- 5-star rating display

Colors:

- Background: Slate 50

10. Upcoming Events

Purpose: Promote conferences and workshops

Layout: 3-column grid with date badge cards

Background: Dark navy with diagonal stripe pattern

Content (3 Events):

- AI & Machine Learning Summit 2026 (Jun 15)

Features:

- Large date badge (month + day)

Colors:

- Background: Dark navy

11. Director Quote & Vision

Purpose: Leadership message and institutional values

Layout: 2-column split (quote left, value cards right)

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Background: Light gradient with decorative circles

Content:

- Left: Director's quote with photo credit

Features:

- Large left border on quote

Colors:

- Background: Slate 100/White
-

12. News & Events Section

Purpose: Latest institutional updates

Layout: Asymmetric grid (1 featured + 3 small)

Background: Cyan/blue gradient with diagonal accent

Content:

- 1 large featured news card (left, 3-col span)

Features:

- Image hover zoom

Colors:

- Background: Gradient cyan-50/primary-50
-

13. Campus Gallery

Purpose: Visual tour of facilities

Layout: 2x3 grid (mobile 2 cols, desktop 3 cols)

Background: White with subtle grid pattern

Content (6 Photos):

- Research Labs

Features:

- 4:3 aspect ratio images

Colors:

- Background: White
-

14. Collaboration Section (Partners)

Purpose: Show industry partnerships

Layout: Infinite horizontal marquee

Background: Dark navy with radial dot pattern

Content:

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- 9 partner logos (IEEE, Huawei, MIT, UET, HEC, IGNITE, Rescue 1122, Punjab Govt, Sports Board)

Features:

- Infinite marquee scroll (25s duration, 20s on mobile)

Colors:

- Background: Dark navy gradient (`#081120`)

Technical:

- Logos duplicated 2x for seamless loop
-

15. Map & Contact Section

Purpose: Location and contact information

Layout: 2-column split (contact cards left, map right)

Background: Light gray with decorative circles

Content:

- Left: 3 contact cards
- Right: Google Maps embed

Features:

- Glassmorphism contact cards

Colors:

- Background: Slate 50/White

Quick Actions:

- Contact Form
-

16. CTA Banner (Final Call-to-Action)

Purpose: Drive conversions before footer

Layout: Centered content with gradient background

Background: Dark gradient with animated blobs

Content:

- Floating "Join KICS Today" badge

Features:

- 3 animated gradient orbs (blob animation)

Colors:

- • • • Secondary button: White/10 glassmorphism
-

Design System & Components

Component Architecture

src/

Reusable Component Patterns

1. Card Component (`.card`)

Properties:

2. Button System

Primary Button (`.btn-primary`)

Background: Primary-600 (#2563eb)

Outline Button (`.btn-outline`)

Border: 2px solid Primary-600

Glass Button

Background: rgba(255, 255, 255, 0.1)

3. Section Labels

Eyebrow (`.eyebrow`)

Text: Primary-600

Section Title (`.section-title`)

Font-weight: 700

Divider (`.divider`)

Width: 3.5rem

Typography System

Font Stack

Primary Font: Inter

Font Sizes

Element	Mobile	Tablet	Desktop
----- ----- ----- -----			
H1	1.75rem (28px)	2.25rem (36px)	3rem (48px)
H2	1.5rem (24px)	2rem (32px)	2.5rem (40px)
H3	1.25rem (20px)	1.5rem (24px)	1.75rem (28px)
Body	0.9375rem (15px)	1rem (16px)	1rem (16px)

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Small	0.875rem (14px)	0.875rem (14px)	0.875rem (14px)
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Font Weights

Name	Weight	Usage
----- ----- -----		
Regular	400	Body text
Medium	500	Subheadings
Semibold	600	UI labels
Bold	700	Headings
Extrabold	800	Hero titles

Line Heights

- Headings: 1.2

Animation System

Scroll Reveal Animations

All sections use AnimateOnScroll component with these variants:

Animation	Effect	Duration	Easing
----- ----- -----			
`reveal`	Fade in + slide up 35px	0.75s	cubic-bezier(.16,1,.3,1)
`reveal-left`	Fade in + slide from left	0.75s	cubic-bezier(.16,1,.3,1)
`reveal-right`	Fade in + slide from right	0.75s	cubic-bezier(.16,1,.3,1)
`reveal-scale`	Fade in + scale from 90%	0.6s	ease

Hover Animations

Card Hover:

```
Transform: translateY(-6px) scale(1.02)
```

Image Zoom:

```
Transform: scale(1.1)
```

Button Hover:

```
Transform: translateY(-2px)
```

Continuous Animations

Animation	Duration	Usage
----- ----- -----		
`marquee`	40s	News ticker

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`marquee-slow`	25s (desktop), 20s (mobile)	Partner logos
`blob`	7s	Floating orbs
`pulse-slow`	3s	Subtle breathing
`float`	3s	Vertical floating
`shimmer`	2s	Loading states
`gradient-shift`	8s	Background gradients

Stagger Delays

Children elements animate with incremental delays:

```
delay={index * 80} // 80ms per item
```

Responsive Design

Breakpoints

Name	Size	Usage
----- ----- -----		
Mobile Small	< 375px	Compact layout
Mobile	< 480px	Single column
Mobile Large	< 768px	1-2 columns
Tablet	768px - 992px	2-3 columns
Desktop	...	Full layout
Large Desktop	...	Max-width containers

Grid Behavior

Stats Section:

- Mobile: 2 columns

Research Cards:

- Mobile: 1 column

Partnership Stats:

- Mobile: 1 column (stacked)

Mobile Optimizations

-
-
-
-
-
-
-

Performance Optimizations

Code Splitting

- `React.memo()` on all section components

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Image Optimization

<img

Animation Performance

will-change: transform; // GPU acceleration

CSS Optimization

- Tailwind Purge: Removes unused CSS

Bundle Size

- React: ~42KB (gzipped)

SEO Implementation

Meta Tags (Every Page)

<title>Page Title | KICS UET Lahore</title>

Structured Data (JSON-LD)

{

Semantic HTML

- `<header>`, `<nav>`, `<main>`, `<section>`, `<article>`, `<footer>`

Key Metrics & Stats

Content Statistics

- Total Sections: 15

Performance Targets

Metric	Target	Status
----- ----- -----		
LCP (Largest Contentful Paint)	< 2.5s	Achieved
FID (First Input Delay)	< 100ms	Achieved
CLS (Cumulative Layout Shift)	< 0.1	Achieved
Lighthouse Score	> 90	Target

Design Philosophy

Principles

Inspiration

-
-
-
-

File Structure

```
kics-frontend-modern/
```

Deployment

Vercel Configuration

- Framework: Vite

Auto-Deployment

- Push to `main` branch on GitHub

Build Time

- Average: ~45 seconds

Support & Maintenance

Browser Support

- Chrome/Edge 90+

Testing Checklist

- Desktop (1920x1080, 1366x768)

Future Enhancements

Planned Features

-
-
-
-
-
-
-

Technical Debt

- Migrate to TypeScript

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Summary

What Makes This Homepage Great

15 comprehensive sections covering all aspects

Modern design with glassmorphism, gradients, animations

Final Verdict

Documentation Generated: May 16, 2026

Version: 2.0.0

Contact: info@kics.edu.pk

Website: <https://kics-frontend-modern.vercel.app>

This documentation covers every aspect of the KICS homepage. For component-level details, refer to the source code in `/src/pages/Home.jsx`.